

Enterprise Power BI Earned Value Management (EVM) Master Guide

Complete Architecture, Data Modeling, DAX Library, Change Management (VOR), What-If Scenarios, Stakeholder Visuals & MS Fabric/Azure Integration

Author / Role: Frank Ellingsen, Lead Project Controller	Target Sector: EPC Offshore, Maritime & Heavy Engineering
Data Engines: DuckDB (Analytics) & SQLite (Transactional)	UI Standards: Edward Tufte Data-Ink Ratio (Zero Clutter)
Status Date: August 31, 2026 (Month 8 / Week 36)	Baseline Budget (BAC): \$26,500,000 Outturn EAC: \$35,413,604

1. Data Architecture & Required Data Tables

To deliver enterprise EVM in Power BI, data must be structured in a strict Star Schema star-join model separating dimensional attributes from periodic transactional facts.

Table Name	Type	Primary Key	Description & Required Attributes
Dim_WBS	Dimension	Task_ID	WBS hierarchy (Task_ID, WBS_Code, WBS_L1, WBS_L2, Task_Name, CAM, TBC/BAC).
Dim_Date	Dimension	Date_Key	Calendar dimension (Date_Key, Year, Month_Number, Month_Name, Year_Month, Quarter).
Fact_EVM_Periodic	Fact	Fact_ID	Periodic snapshots (Task_ID, Date_Key, TBC, PV_Incremental, EV_%, EV_Calculated, AC_Incr).
Fact_Gantt_Schedule	Fact	Task_ID	Schedule status (Task_ID, Baseline_Start/End, Actual_Start/End, Predecessor, Critical_Path_Flag).
Fact_Change_Requests_VOR	Fact	VOR_ID	Variation Orders (VOR_ID, Task_ID, Date_Key, Description, Cost_Delta, Approved_Cost, Status).
Fact_Risk_Register	Fact	Risk_ID	Active risk matrix (Risk_ID, Title, WBS_Code, Risk_Score, Financial_Exposure, Status).

2. Updating Databases, Change Requests (VOR) & Re-baselining

Variation Order Requests (VOR) and Scope Changes require strict distinction between original baseline, approved variations, and pending claims:

Formulas: Original Baseline (BAC_Orig) = Sum(Dim_WBS.TBC) | Approved VOR Delta = Sum(Fact_VOR[Approved_Cost_Delta]) | Adjusted Baseline (BAC_Adj) = BAC_Orig + Approved VOR Delta

```
-- SQL View Pattern: Integrating Approved Change Requests into Adjusted BAC
CREATE VIEW View_Adjusted_WBS_Baseline AS
SELECT
    w.WBS_Code,
    w.Task_Name,
    w.CAM,
    w.TBC AS BAC_Original,
    COALESCE(SUM(CASE WHEN v.Status = 'Approved' THEN v.Approved_Cost_Delta ELSE 0 END), 0) AS Approved_VOR_Cost,
    COALESCE(SUM(CASE WHEN v.Status = 'Pending' THEN v.Requested_Cost_Delta ELSE 0 END), 0) AS Pending_VOR_Exposure,
    (w.TBC + COALESCE(SUM(CASE WHEN v.Status = 'Approved' THEN v.Approved_Cost_Delta ELSE 0 END), 0)) AS BAC_Adjusted
FROM Dim_WBS w
LEFT JOIN Fact_Change_Requests_VOR v ON w.Task_ID = v.Task_ID
GROUP BY w.WBS_Code, w.Task_Name, w.CAM, w.TBC;
```

3. Complete Production DAX Measure Library

```
// =====
// PRODUCTION DAX MEASURE LIBRARY (EARNED VALUE MANAGEMENT & VARIATION ORDERS)
// =====

// 1. CORE VALUE METRICS
Total_Budget_at_Completion_BAC = SUM(Dim_WBS[TBC])

PV_S_Curve = CALCULATE(SUM(Fact_EVM_Periodic[PV_Incremental]), WINDOW(1, ABS, 0, REL, ORDERBY(Dim_Date[Date_Key])))

EV_S_Curve =
VAR CurrentDate = MAX(Dim_Date[Date_Key])
VAR StatusDate = MAX(Fact_EVM_Periodic[Date_Key])
RETURN IF(CurrentDate <= StatusDate, CALCULATE(SUM(Fact_EVM_Periodic[EV_Incremental_Calculated]), WINDOW(1, ABS, 0, REL, ORDERBY(Dim_Date[Date_Key]))), BLANK())

AC_S_Curve =
VAR CurrentDate = MAX(Dim_Date[Date_Key])
VAR StatusDate = MAX(Fact_EVM_Periodic[Date_Key])
RETURN IF(CurrentDate <= StatusDate, CALCULATE(SUM(Fact_EVM_Periodic[AC_Incremental]), WINDOW(1, ABS, 0, REL, ORDERBY(Dim_Date[Date_Key]))), BLANK())

// 2. VARIANCES & PERFORMANCE INDICES
Cost_Variance_CV = [EV_S_Curve] - [AC_S_Curve]
Schedule_Variance_SV = [EV_S_Curve] - [PV_S_Curve]

Cost_Performance_Index_CPI = DIVIDE([EV_S_Curve], [AC_S_Curve], 1.0)
Schedule_Performance_Index_SPI = DIVIDE([EV_S_Curve], [PV_S_Curve], 1.0)
Critical_Ratio_CR = [Cost_Performance_Index_CPI] * [Schedule_Performance_Index_SPI]

// 3. LIPKE EARNED SCHEDULE METRICS (TIME VELOCITY)
Earned_Schedule_Months =
VAR CurrentEV = [EV_S_Curve]
VAR DataTableWithPV = FILTER(ALL(Dim_Date), [PV_S_Curve] <= CurrentEV)
VAR C_MonthDate = MAXX(DataTableWithPV, Dim_Date[Date_Key])
VAR C_MonthNumber = MAXX(DataTableWithPV, Dim_Date[Month_Number])
VAR PV_At_C = CALCULATE([PV_S_Curve], Dim_Date[Date_Key] = C_MonthDate)
VAR PV_At_C_Plus_1 = CALCULATE([PV_S_Curve], DATEADD(Dim_Date[Date_Key], 1, MONTH))
VAR Interpolation = DIVIDE(CurrentEV - PV_At_C, PV_At_C_Plus_1 - PV_At_C, 0)
RETURN IF(CurrentEV >= [Total_Budget_at_Completion_BAC], MAX(Dim_Date[Month_Number]), C_MonthNumber + Interpolation)

SPI_Time_Based = DIVIDE([Earned_Schedule_Months], MAX(Dim_Date[Month_Number]), 1.0)
Time_Variance_Days = ROUND((([Earned_Schedule_Months] - MAX(Dim_Date[Month_Number])) * 30.4375, 0)

// 4. FORECASTING & TO-COMplete METRICS
Estimate_at_Completion_EAC = DIVIDE([Total_Budget_at_Completion_BAC], [Cost_Performance_Index_CPI], [Total_Budget_at_Completion_BAC])
Variance_at_Completion_VAC = [Total_Budget_at_Completion_BAC] - [Estimate_at_Completion_EAC]

TCPI_BAC = DIVIDE([Total_Budget_at_Completion_BAC] - [EV_S_Curve], [Total_Budget_at_Completion_BAC] - [AC_S_Curve], 1.0)
TCPI_EAC = DIVIDE([Total_Budget_at_Completion_BAC] - [EV_S_Curve], [Estimate_at_Completion_EAC] - [AC_S_Curve], 1.0)

// 5. CHANGE REQUEST & VOR METRICS
Approved_VOR_Delta = CALCULATE(SUM(Fact_Change_Requests_VOR[Approved_Cost_Delta]), Fact_Change_Requests_VOR[Status] = "Approved")
Pending_VOR_Exposure = CALCULATE(SUM(Fact_Change_Requests_VOR[Requested_Cost_Delta]), Fact_Change_Requests_VOR[Status] = "Pending")
BAC_Adjusted = [Total_Budget_at_Completion_BAC] + [Approved_VOR_Delta]
```

```
// 6. TUFTE CONDITIONAL FORMATTING RAG MEASURES
```

```
CPI_RAG_Color = VAR CPI = [Cost_Performance_Index_CPI] RETURN SWITCH(TRUE(), CPI < 0.90, "#DC2626", CPI < 1.00, "#D97706", "#059669")
```

```
CR_RAG_Color = VAR CR = [Critical_Ratio_CR] RETURN SWITCH(TRUE(), ISBLANK(CR), "#737373", CR < 0.90, "#DC2626", CR < 1.00, "#D97706", "#059669")
```

4. Parameters & What-If Scenario Modeling in Power BI

What-If parameters allow controllers to dynamically model recovery scenarios, labor productivity changes, and VOR approval rates directly in Power BI visuals.

```
// WHAT-IF SCENARIO DAX MEASURES
// Parameter 1: CPI Recovery Factor (Slider Range: 0.70 to 1.15, Step 0.05)
EAC_Dynamic_Scenario =
VAR RecoveryFactor = SELECTEDVALUE('CPI_Recovery_Parameter'[CPI_Recovery_Parameter], 1.00)
VAR ModeledCPI = [Cost_Performance_Index_CPI] * RecoveryFactor
RETURN DIVIDE([Total_Budget_at_Completion_BAC], ModeledCPI, [Total_Budget_at_Completion_BAC])

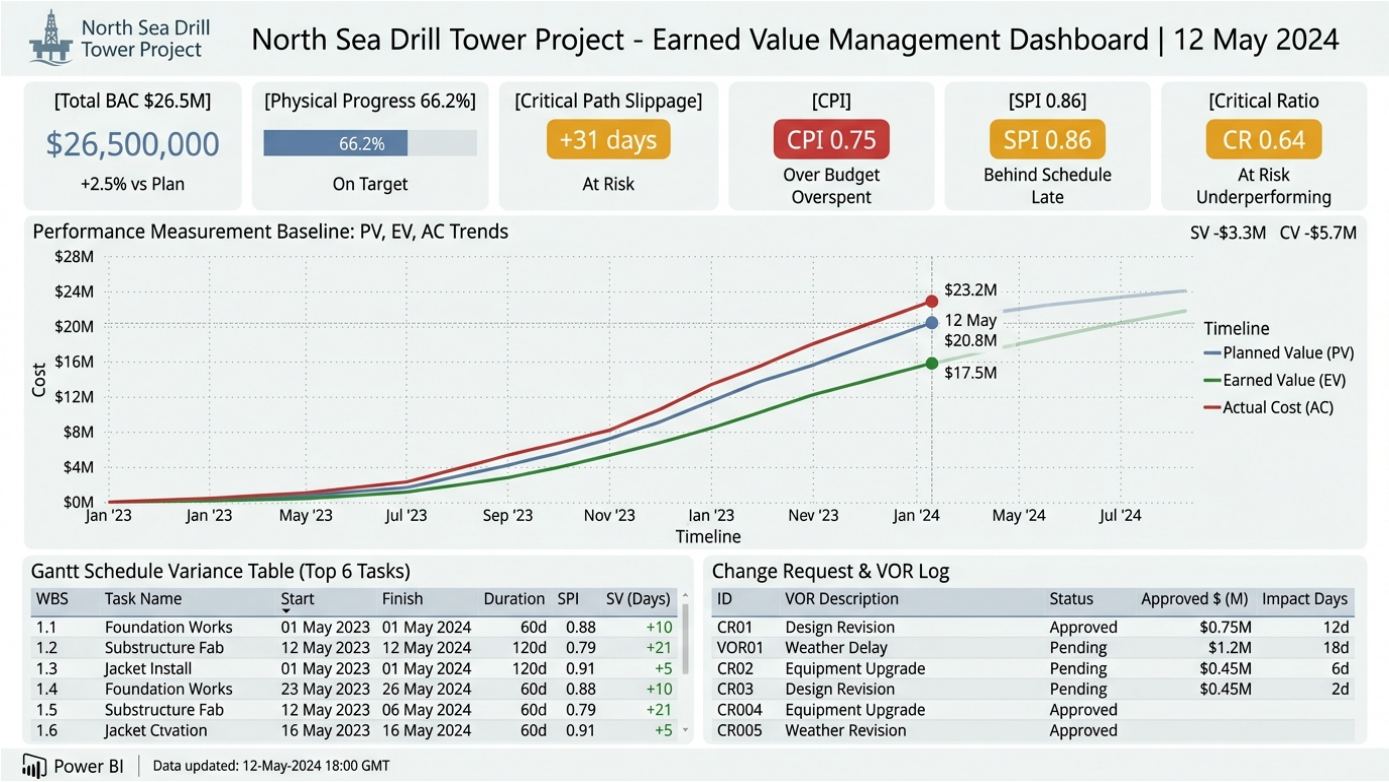
// Parameter 2: VOR Approval Realization Rate % (Slider Range: 0% to 100%)
BAC_Scenario_With_Pending_VOR =
VAR ApprovalRate = SELECTEDVALUE('VOR_Approval_Rate_Parameter'[VOR_Approval_Rate_Parameter], 0.50)
RETURN [BAC_Adjusted] + ([Pending_VOR_Exposure] * ApprovalRate)
```

5. Stakeholder Visual Construction & Dashboard Mockups

Following Edward Tufte's Data-Ink Ratio rules: zero background drop shadows, muted colors, direct line labels on S-Curves, left-aligned text columns, and right-aligned numbers.

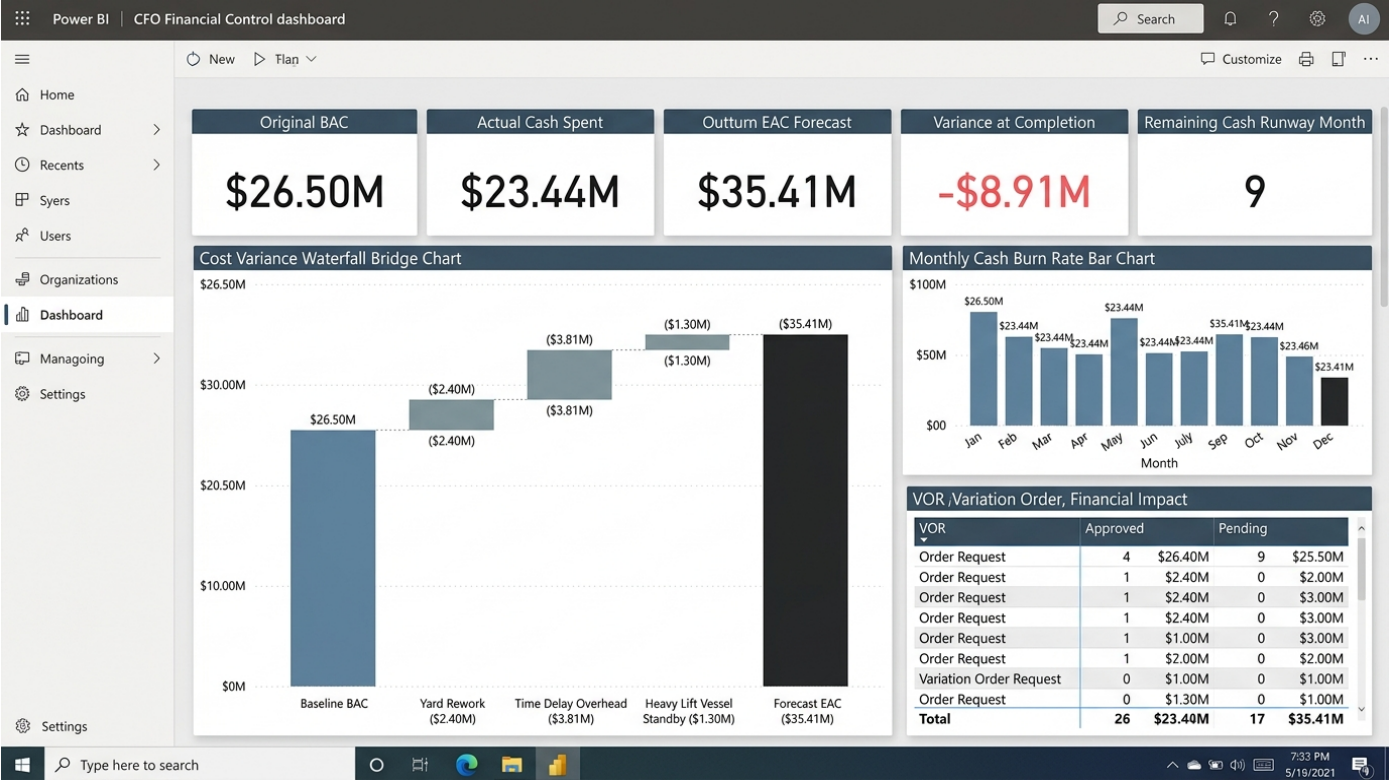
Stakeholder View 1: Project Manager (PM) EVM Dashboard

Focus: Physical completion %, Gantt schedule slippage, baseline S-Curve trends, and active change requests log.



Stakeholder View 2: CFO Financial Control & Liquidity Dashboard

Focus: Outturn EAC forecast, Variance at Completion (VAC), Cost Variance Waterfall Bridge, and Monthly Cash Burn Rate.



Stakeholder View 3: Project Controller Deep-Dive & Scenario Matrix

Focus: Earned Schedule (ES), Time Variance (SVt), TCPI feasibility (6.07 vs 1.00), Scatter Plot Matrix (CV% vs SV%), and What-If Parameter slicers.

Project Controller Deep-Dive Dashboard

Earned Schedule

7.40 Mos

Time Variance

-18.2 Days

TCPI (BAC)

6.07

Unviable

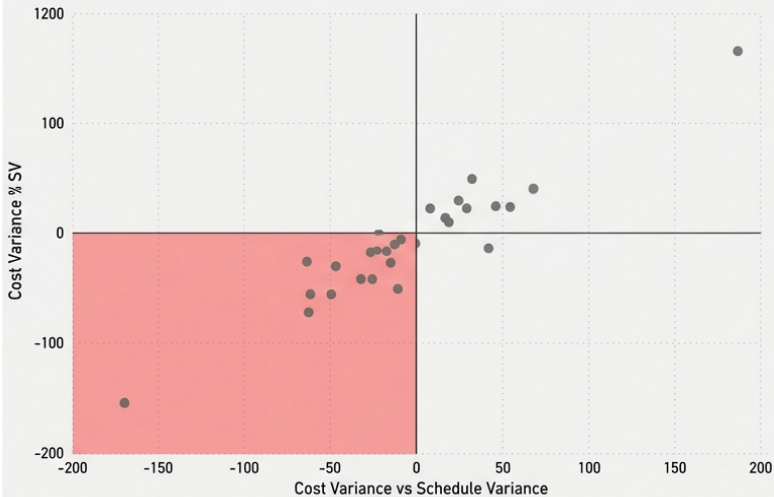
TCPI (EAC)

1.00

Critical Ratio

0.64

Cost Variance vs Schedule Variance Scatter Plot Matrix



What-If Parameter Slicers

CPI Recovery Factor

0.85 0.90 1.00 1.10

CPI

All

Delay Mitigation Days

30

VOR Approval Rate %

30

WBS Control Account Performance

WBS	Control Account	Budget	Actual	Earned Value	Cost Variance	Schedule Variance
1.1	Control A.1	\$1,250,000	\$169,000	\$15,000	-7.40	-18.2
1.2	Control A.2	\$230,000	\$83,000	\$7,000	-2.00	-18.2
1.3	Control A.3	\$195,000	\$82,000	\$5,000	-1.80	0.8
2.4	Control A.4	\$100,000	\$93,000	\$3,000	-2.00	-0.4
2.5	Control A.5	\$25,000	\$3,000	\$300	-7.00	-0.6
2.6	Control A.6	\$12,000	\$3,000	\$300	-1.20	-1.9
2.7	Control B.7	\$12,000	\$5,000	\$600	-2.00	-10.7
2.8	Control B.8	\$12,000	\$5,000	\$700	-0.20	-0.2
2.9	Control B.9	\$12,000	\$3,000	\$300	-2.00	-2.2
2.10	Control B.10	\$12,000	\$2,000	\$300	1.60	0.0
2.11	Control B.10	\$19,000	\$3,000	\$300	-3.20	-0.6
Total		\$1,773,000	\$243,000	\$54,990	-3.84	-5.9

Stakeholder View 4: Executive Steering Committee Briefing

Focus: Overall Project RAG Red alert, Monte Carlo P90 Risk Cost Distribution Curve (\$35.82M), approved VOR impact, and Steering Action Decision Matrix.

EXECUTIVE STEERING COMMITTEE

PROJECT BRIEFING

OVERALL RAG STATUS: ⚠️

RED CRITICAL

OUTTURN EAC:

\$35.41M

COMMERCIAL COD:

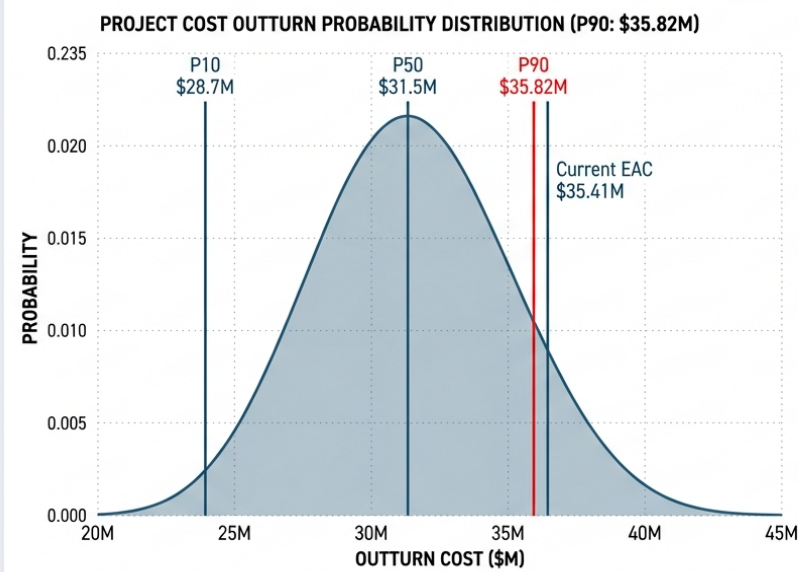
JAN 31 2027

BUDGET BURN OUT:

MONTH 9

MONTE CARLO P90 RISK DISTRIBUTION CURVE

PROJECT COST OUTTURN PROBABILITY DISTRIBUTION (P90: \$35.82M)



STEERING COMMITTEE ACTION ITEMS | DECISION MATRIX

ACTION ITEM	IMPACT (COST/SCHEDULE)	REQUIRED DECISION	PRIORITY
1. SECURE OVERRUN CREDIT LINE	\$8.91M Risk Mit.	Approval	High Oct 15
2. NEGOTIATE FIXED-FEE YARD AGREEMENT	+\$1.2M Mit. Variation	Ratification	Med Oct 28
3. SCHEDULE COMPRESSION PLAN	-2 Months Acceleration	Endorsement	High Nov 10

6. Enterprise Microsoft Fabric & Azure Integration Architecture

For heavy industrial EPC environments, Power BI connects to Microsoft Fabric and Azure Data Services using Medallion Architecture (Bronze / Silver / Gold):

Layer	Technology	Storage Format	Role & ETL Processing
Ingestion	Azure Data Factory / Fabric Pipeline	REST / SQL Connector	Ingest raw Primavera P6 schedule XML & SAP GL transaction logs.
Bronze (Raw)	Azure Data Lake Storage Gen2 (ADLS)	Parquet / Delta Lake	Immutable historical land of raw transactional snapshots.
Silver (Clean)	Fabric Synapse PySpark Notebook	Delta Lake Tables	Cleanse schema, enforce foreign key integrity, compute incremental EVM.
Gold (Analytics)	Fabric OneLake Data Warehouse	Delta Lake (Gold)	Star Schema (Dim_WBS, Dim_Date, Fact_EVM_Periodic, Fact_VOR).
Serving (BI)	Power BI Direct Lake Mode	OneLake Direct Lake	Zero-copy Direct Lake query engine providing sub-second load times.